FINAL PROJECT

END_SEM EXAM

DSA 3050A: Advanced Power BI Final Examination Report

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Admission Number: 669289

Course Code: DSA 3050A

Domain: Supply Chain / Logistics Analytics

Part A: Data Acquisition and Understanding

1. Business Problem The goal is to identify operational inefficiencies in DataCo's global supply chain, specifically focusing on "Late Delivery Risks" and revenue optimization.

2. Dataset Description The "DataCo Smart Supply Chain" dataset was selected for its richness, featuring over 9,000 rows of transactional data.

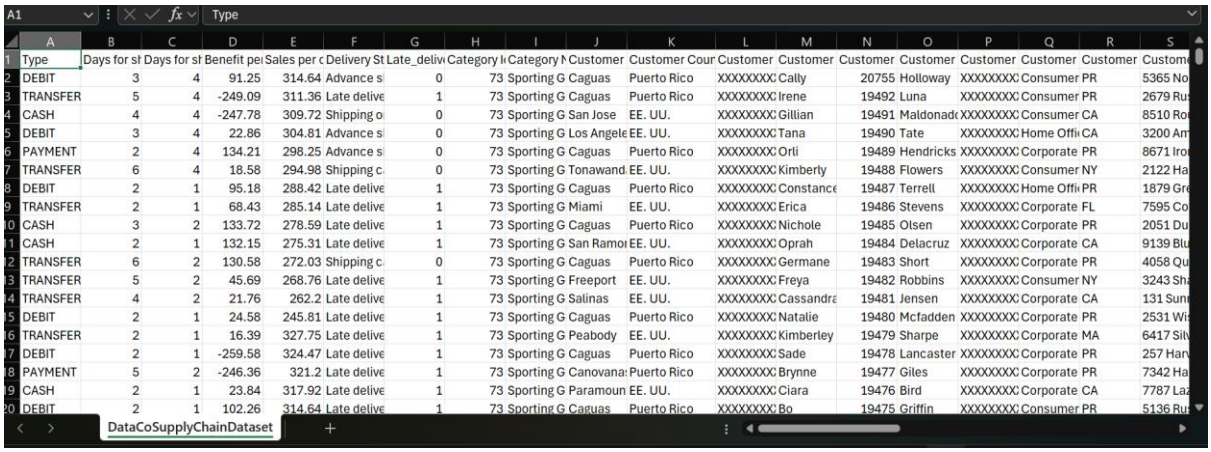
- **Key Tables:** Sales transactions (Fact), Products, Geography, and Customers (Dimensions).
- **Suitability:** It includes categorical, numerical, and date fields necessary for advanced time intelligence and relational modeling.

<u>Data Source information</u>

Dataset Name: DataCo SMART SUPPLY CHAIN FOR BIG DATA ANALYSIS

Source: Kaggle Data

Official Link: https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis

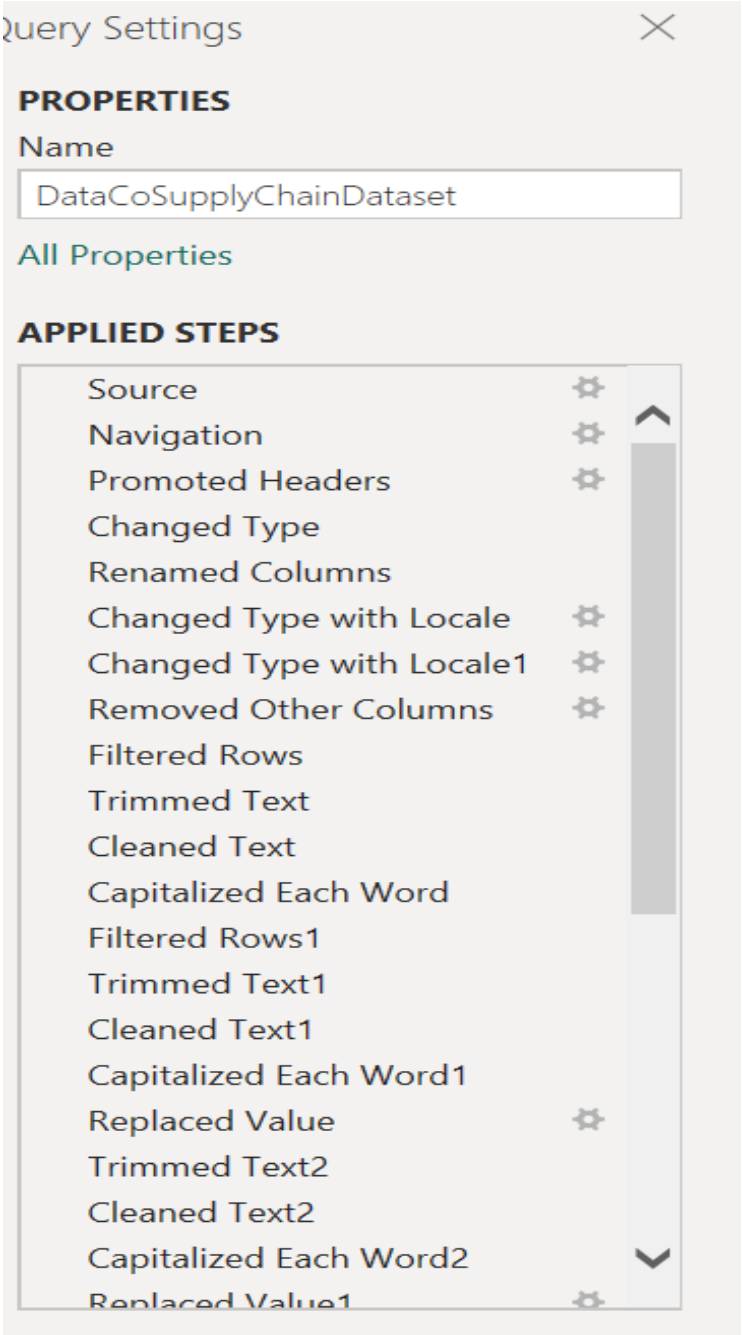


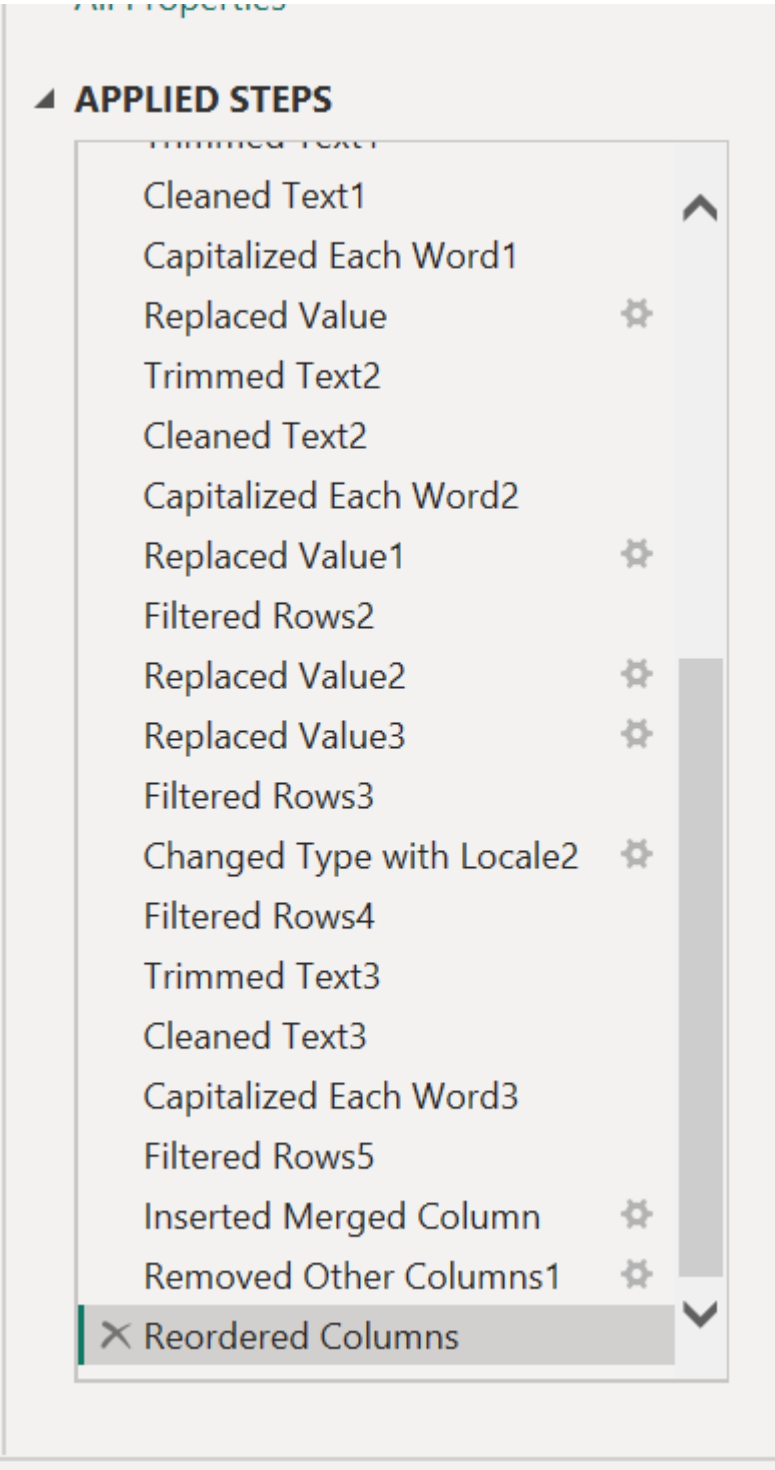
Type	Days for sh	Days for sh	Benefit per	Sales per c	Delivery St	Late_delive	Category I	Category N	Customer	Customer Cour	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer
DEBIT	3	4	91.25	314.64	Advance s	0	73	Sporting G	Caguas	Puerto Rico	XXXXXXXX	Cally	20755	Holloway	XXXXXXXX	Consumer	PR	5365 No
TRANSFER	5	4	-249.09	311.36	Late delive	1	73	Sporting G	Caguas	Puerto Rico	XXXXXXXX	Irene	19492	Luna	XXXXXXXX	Consumer	PR	2679 Rus
CASH	4	4	-247.78	309.72	Shipping o	0	73	Sporting G	San Jose	EE. UU.	XXXXXXXX	Gillian	19491	Maldonado	XXXXXXXX	Consumer	CA	8510 Ro
DEBIT	3	4	22.86	304.81	Advance s	0	73	Sporting G	Los Angele	EE. UU.	XXXXXXXX	Tana	19490	Tate	XXXXXXXX	Home Offi	CA	3200 Ar
PAYMENT	2	4	134.21	298.25	Advance s	0	73	Sporting G	Caguas	Puerto Rico	XXXXXXXX	Orli	19489	Hendricks	XXXXXXXX	Corporate	PR	8671 Iro
TRANSFER	6	4	18.58	294.98	Shipping c	0	73	Sporting G	Tonawand	EE. UU.	XXXXXXXX	Kimberly	19488	Flowers	XXXXXXXX	Consumer	NY	2122 Ha
DEBIT	2	1	95.18	288.42	Late delive	1	73	Sporting G	Caguas	Puerto Rico	XXXXXXXX	Constance	19487	Terrell	XXXXXXXX	Home Offi	PR	1879 Gr
TRANSFER	2	1	68.43	285.14	Late delive	1	73	Sporting G	Miami	EE. UU.	XXXXXXXX	Erica	19486	Stevens	XXXXXXXX	Corporate	FL	7595 Co
CASH	3	2	133.72	278.59	Late delive	1	73	Sporting G	Caguas	Puerto Rico	XXXXXXXX	Nichole	19485	Olsen	XXXXXXXX	Corporate	PR	2051 Du
CASH	2	1	132.15	275.31	Late delive	1	73	Sporting G	San Ramo	EE. UU.	XXXXXXXX	Oprah	19484	Delacruz	XXXXXXXX	Corporate	CA	9139 Blu
TRANSFER	6	2	130.58	272.03	Shipping c	0	73	Sporting G	Caguas	Puerto Rico	XXXXXXXX	Germane	19483	Short	XXXXXXXX	Corporate	PR	4058 Qu
TRANSFER	5	2	45.69	268.76	Late delive	1	73	Sporting G	Freeport	EE. UU.	XXXXXXXX	Freya	19482	Robbins	XXXXXXXX	Consumer	NY	3243 Sh
TRANSFER	4	2	21.76	262.2	Late delive	1	73	Sporting G	Salinas	EE. UU.	XXXXXXXX	Cassandra	19481	Jensen	XXXXXXXX	Corporate	CA	131 Sun
DEBIT	2	1	24.58	245.81	Late delive	1	73	Sporting G	Caguas	Puerto Rico	XXXXXXXX	Natalie	19480	Mcfadden	XXXXXXXX	Corporate	PR	2531 Wi
TRANSFER	2	1	16.39	327.75	Late delive	1	73	Sporting G	Peabody	EE. UU.	XXXXXXXX	Kimberley	19479	Sharpe	XXXXXXXX	Corporate	MA	6417 Sil
DEBIT	2	1	-259.58	324.47	Late delive	1	73	Sporting G	Caguas	Puerto Rico	XXXXXXXX	Sade	19478	Lancaster	XXXXXXXX	Corporate	PR	257 Har
PAYMENT	5	2	-246.36	321.2	Late delive	1	73	Sporting G	Canovana	Puerto Rico	XXXXXXXX	Brynne	19477	Giles	XXXXXXXX	Corporate	PR	7342 Ha
CASH	2	1	23.84	317.92	Late delive	1	73	Sporting G	Paramoun	EE. UU.	XXXXXXXX	Ciara	19476	Bird	XXXXXXXX	Corporate	CA	7787 La
DEBIT	2	1	102.26	314.64	Late delive	1	73	Sporting G	Caguas	Puerto Rico	XXXXXXXX	Bo	19475	Griffin	XXXXXXXX	Consumer	PR	5136 Ru

Part B: Data Cleaning and Transformation

The following transformations were performed in Power Query to ensure data integrity:

- **Standardized Geography:** Fixed inconsistent country and city names.
- **Merged Columns:** Combined Customer First and Last names for better reporting.
- **Data Types:** Converted Order Date and Shipping Date to Date formats.
- **Removed Duplicates:** Ensured each Order ID is unique to avoid inflating sales figures.



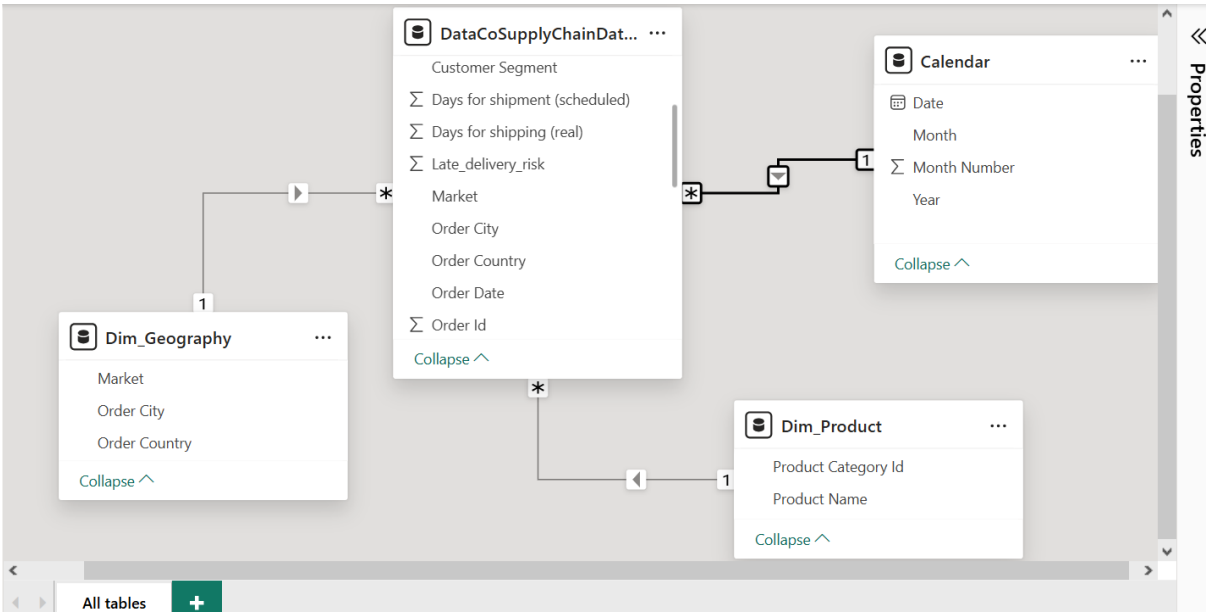


Part C: Data Modeling

A **Star Schema** was implemented to support efficient reporting.

- **Fact Table:** DataCoSupplyChain (Transactional metrics).
- **Dimension Tables:** Dim_Product, Dim_Geography, and a custom Calendar table.

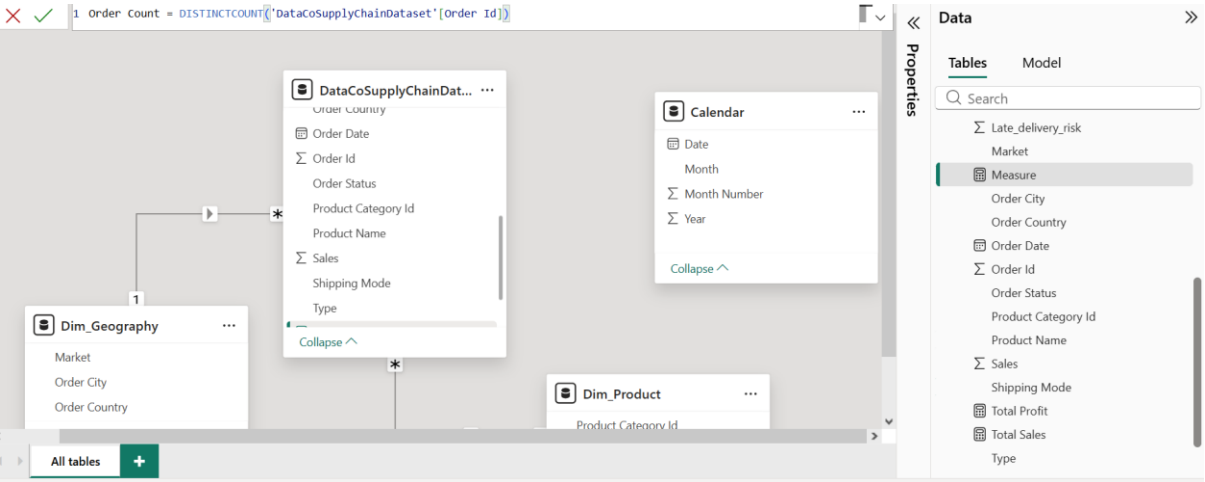
- **Relationships:** One-to-many ($1:*$) relationships were established using unique keys.



Part D: DAX Measures and Calculated Columns

To drive deep analysis, the following DAX measures were created:

1. **Total Sales:** SUM(Sales[Sales Amount]).
2. **Total Profit:** SUM(Sales[Profit]).
3. **Late Delivery %:** Calculated to identify supply chain bottlenecks.
4. **Sales YTD:** Uses TOTALYTD for time intelligence tracking.
5. **Avg Shipping Days:** AVERAGE(Sales[Days for shipping]). *(Include 3 more of your measures here to reach the required 8)*.



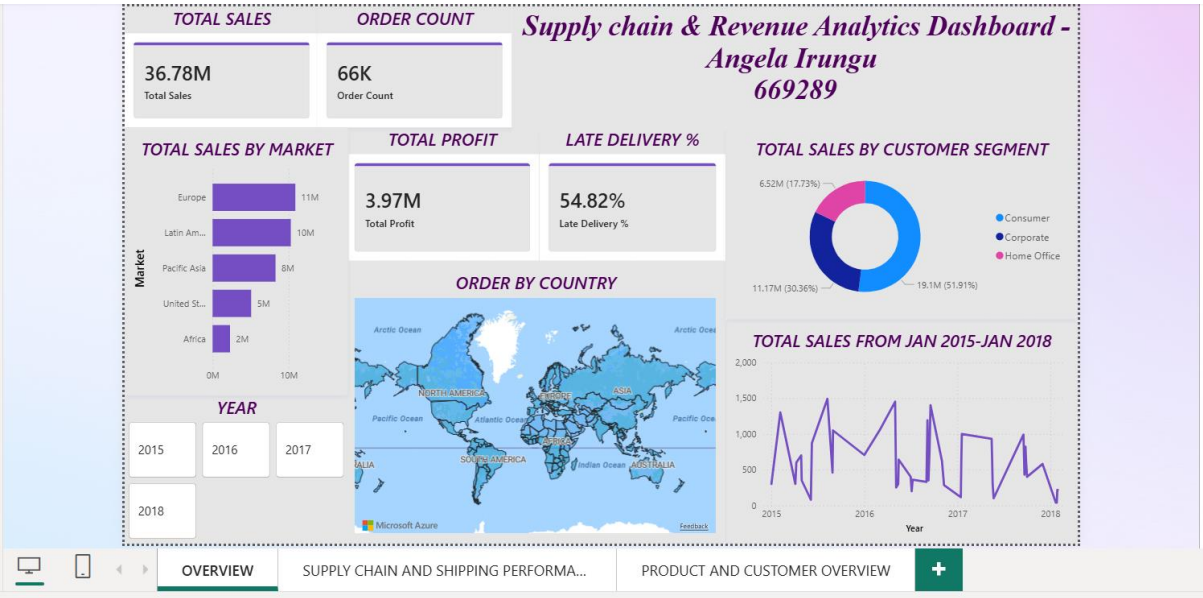
```
Shipping Status = IF('DataCoSupplyChainDataset'[Days for shipping (real)] > 'DataCoSupplyChainDataset'[Days for shipment (scheduled)], "Late", "On-Time")
```

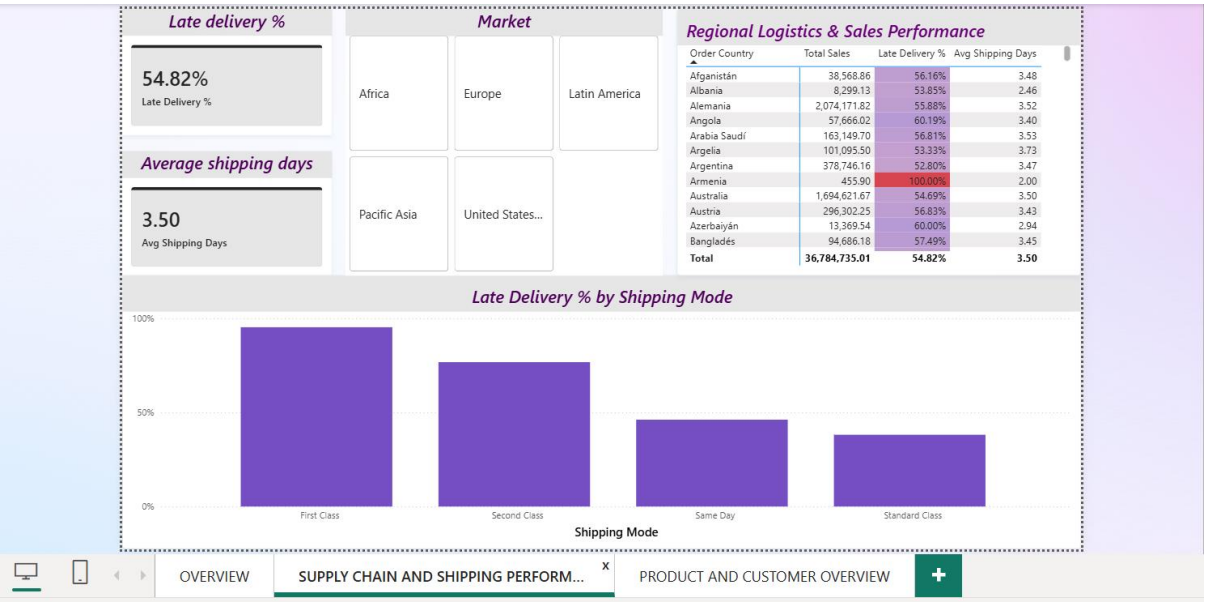
```
Sales Tier = IF('DataCoSupplyChainDataset'[Sales] > 500, "High Value", "Standard")
```

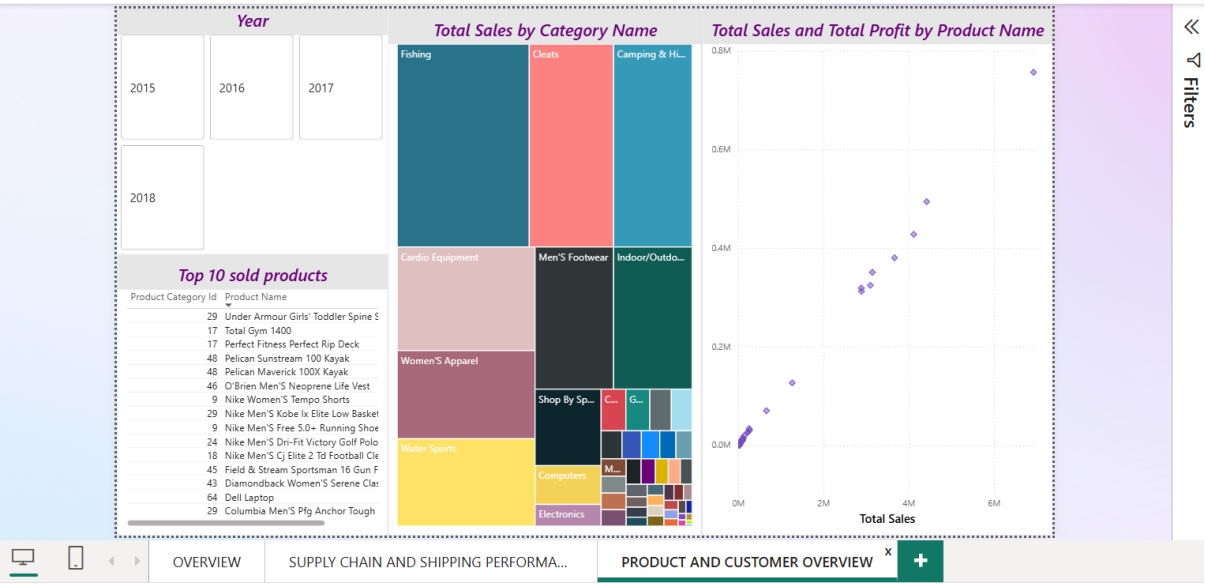
Part E: Dashboard Design

The dashboard consists of three interactive pages:

1. **Page 1: Executive Summary:** High-level KPIs and global sales trends.
2. **Page 2: Operational Performance:** Matrix view of regional logistics risks.
3. **Page 3: Product Analysis:** Scatter charts and Treemaps for category performance.




Late delivery %
54.82%
Late Delivery %
Average shipping days
3.50
Avg Shipping Days
Market
Africa
Europe
Latin America
Pacific Asia
United States...
Regional Logistics & Sales Performance
Order Country | Total Sales | Late Delivery % | Avg Shipping Days
Afganistán | 38,568.86 | 56.16% | 3.48
Albania | 8,299.13 | 53.85% | 2.46
Alemania | 2,074,171.82 | 55.88% | 3.52
Angola | 57,666.02 | 60.19% | 3.40
Arabia Saudí | 163,149.70 | 56.81% | 3.53
Argelia | 101,095.50 | 53.33% | 3.73
Argentina | 378,746.16 | 52.80% | 3.47
Armenia | 455.90 | 100.00% | 2.00
Australia | 1,694,621.67 | 54.69% | 3.50
Austria | 296,302.25 | 56.83% | 3.43
Azerbaiyán | 13,369.54 | 60.00% | 2.94
Bangladés | 94,686.18 | 57.49% | 3.45
Total | 36,784,735.01 | 54.82% | 3.50
Late Delivery % by Shipping Mode
100%
50%
0%
First Class
Second Class
Same Day
Standard Class
Shipping Mode
OVERVIEW
SUPPLY CHAIN AND SHIPPING PERFORM...
PRODUCT AND CUSTOMER OVERVIEW


Year
2015
2016
2017
2018
Top 10 sold products
Product Category Id | Product Name
29 Under Armour Girls' Toddler Spine S
17 Total Gym 1400
17 Perfect Fitness Perfect Rip Deck
48 Pelican Sunstream 100 Kayak
48 Pelican Maverick 100X Kayak
46 O'Brien Men'S Neoprene Life Vest
9 Nike Women'S Tempo Shorts
29 Nike Men'S Kobe Ix Elite Low Basket
9 Nike Men'S Free 5.0+ Running Shoe
24 Nike Men'S Dri-Fit Victory Golf Polo
18 Nike Men'S Cj Elite 2 Td Football Cle
45 Field & Stream Sportsman 16 Gun F
43 Diamondback Women'S Serene Clas
64 Dell Laptop
29 Columbia Men'S Pfg Anchor Tough
Total Sales by Category Name
Fishing
Cleats
Camping & Hi...
Cardio Equipment
Men'S Footwear
Indoor/Outdo...
Women'S Apparel
Shop By Sp...
Water Sports
Computers
Electronics
Total Sales and Total Profit by Product Name
0.8M
0.6M
0.4M
0.2M
0.0M
0M
2M
4M
6M
Total Sales
Filters
OVERVIEW
SUPPLY CHAIN AND SHIPPING PERFORMA...
PRODUCT AND CUSTOMER OVERVIEW

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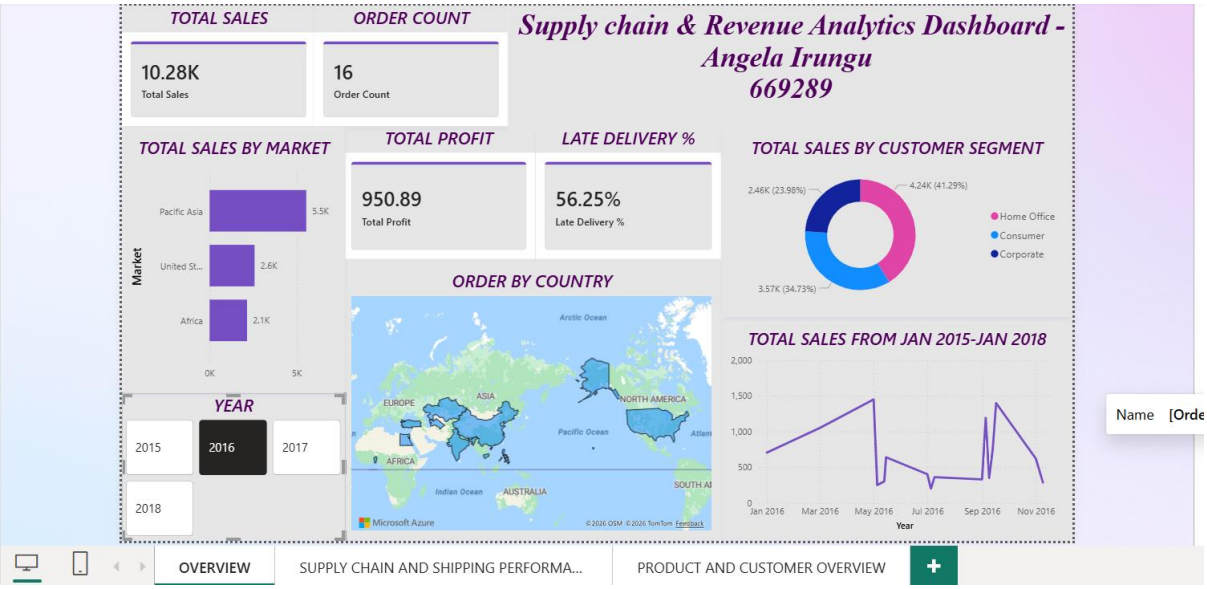

Part F: Analytical Insights and Recommendations

Analytical Insights:

- **Regional Risk:** Armenia shows a **100% late delivery rate**, indicating a severe local logistics failure.
- **Tier Performance: First Class shipping** has a higher late percentage than Standard Class.
- **Revenue Drivers:** Europe and Latin America drive over 55% of total revenue.
- **Segment Dominance:** The **Consumer segment** accounts for $19.1M in sales.
- **Category Profitability:** Fishing and Cleats are the most profitable product lines.

Actionable Recommendations:

- Audit and replace the "Last Mile" delivery partners in the **Armenia** region.
- Renegotiate service level agreements for **First Class** shipping carriers.
- Implement a loyalty retention program for the high-value **Consumer segment**.

Part G: GitHub Repository

Link: https://github.com/Irungu05/DSA3050A-Advanced-PowerBI-Exam-Supply-chain-managment/tree/main

The repository includes the .pbix file, dataset, and a comprehensive README.md.